Context-Aware Question Answering for Vietnamese University Admissions via Multi-LLM Architecture

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2025 International Conference on Multimedia Analysis and Pattern Recognition (MAPR) | 979-8-3315-5466-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/MAPR67746.2025.11133885

***Abstract*—The increasing demand for transparent and accessible communication in university admissions highlights the need for intelligent systems to support students, especially in complex contexts like Vietnam. Traditional QA approaches, primarily based on rule-based systems and basic retrieval models, face scalability and contextual understanding challenges, and current Vietnamese benchmarks do not fully address admissions-specific needs. To overcome these gaps, we introduce a contextual chunking framework and a modular multi-LLM architecture to improve question answering performance over long, complex admission documents. Our framework addresses key limitations in semantic preservation and task modularity, enabling scalable educational chatbot systems. Experiments conducted on a newly constructed multimodal Vietnamese admissions dataset validate the effectiveness of our methods, achieving substantial gains in retrieval accuracy and answer generation quality compared to baseline approaches. These results demonstrate the promise of our approach for advancing intelligent educational QA systems in low-resource, policy-intensive domains.**

***Index Terms*—University Admissions, question answering systems, contextual chunking**

I. Introduction

The increasing demand for transparent, accessible, and responsive communication in the field of university admissions has underscored the necessity of intelligent systems that can support prospective students in obtaining accurate information. In Vietnam, where hundreds of thousands of high school graduates participate in the national university entrance process each year, timely and reliable access to admissions-related details, such as entry conditions, academic programs, tuition policies, and scholarship opportunities, is critical. However, such information is often scattered across various sources and presented in formats that are not easily searchable, posing significant challenges for students attempting to navigate the admissions process.

Traditional approaches to question answering (QA) [5] in the domain of university admissions have primarily relied on rule-based systems and basic retrieval models. These systems are typically constructed using manually curated templates or keyword-based search algorithms trained on small-scale or domain-agnostic datasets. While such approaches provide baseline functionality, they exhibit notable limitations in scalability, contextual understanding, and adaptability to nuanced or ambiguous queries. Furthermore, although Vietnamese-language benchmarks have advanced general-purpose and visual question answering capabilities, they are not specifically tailored to capture the unique information structures and interaction patterns present in university admissions queries, such as eligibility rules, application workflows, or multi-year academic program configurations.

To address these limitations, this study presents three core contributions that collectively advance the state of question answering systems in the Vietnamese educational domain. First, we introduce a novel and domain-specific QA dataset [12] compiled from diverse, institutionally validated sources, including official university websites, admission handbooks, academic brochures, regulatory documents, and infographic materials. This dataset spans multiple modalities – structured tables, narrative texts, and visual materials – and is systematically categorized into major thematic domains such as enrollment pathways, entrance benchmarks, tuition and fee structures, program descriptions, and scholarship criteria. We provide comprehensive statistics to characterize the dataset, highlighting frequently asked questions and topic distributions, thereby establishing a benchmark for future research.

Second, we address the challenge of processing lengthy and heterogeneous documents – a common characteristic of admission materials – by proposing a contextual chunking framework. Unlike traditional chunking strategies [14] that apply fixed-length or naive sentence-based segmentation, our method leverages semantic signals to generate context-preserving text segments. This approach is motivated by recent work on semantic-aware segmentation for document retrieval [1], [13], ensuring that each chunk retains meaningful connections to the original discourse structure. By maintaining discourse coherence, our approach enhances retrieval precision and improves downstream QA performance, as demonstrated through evaluations against standard baselines on long-document processing tasks.

Third, we design and implement a modular multi-LLM-based architecture in which four specialized Large Language Models (LLMs) are assigned distinct roles: (1) query understanding, (2) historical and contextual expansion, (3) document retrieval, and (4) answer generation. This design contrasts

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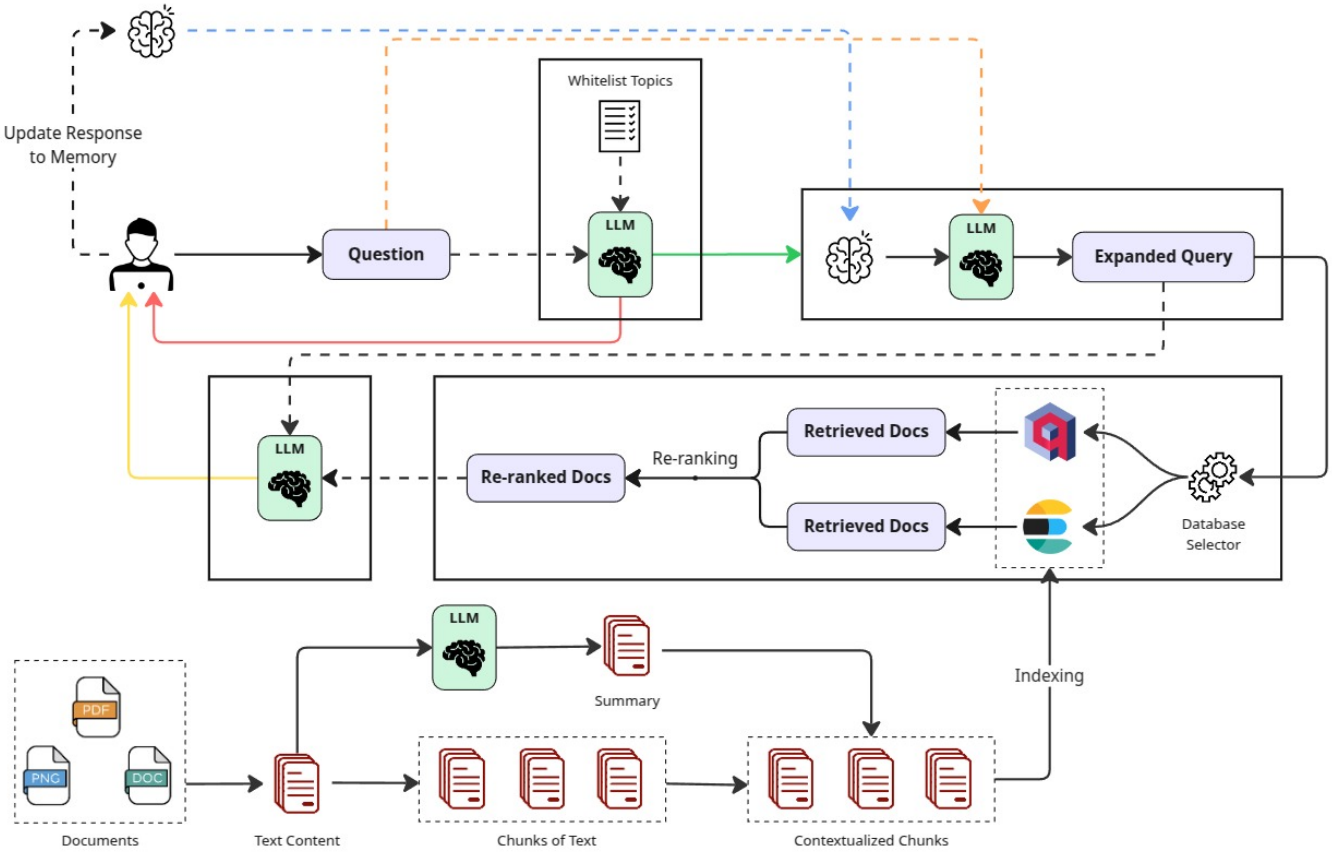

Figure 1. Overview of the proposed modular system pipeline for university admissions QA, comprising contextual chunking for semantic-preserving document segmentation and a multi-LLM architecture for query understanding, context enrichment, document retrieval, and grounded response generation.

with monolithic dialogue systems that delegate all tasks to a single model, often resulting in decreased transparency and diminished task-specific performance. By separating responsibilities across multiple agents, our architecture enables greater modularity, facilitates performance diagnostics, and allows for targeted improvements in each subcomponent.

In conclusion, this paper contributes a comprehensive dataset, a novel context-aware document processing methodology, and a scalable LLM-based system architecture for university admissions QA in Vietnamese. These contributions not only address key limitations in current approaches but also lay a robust foundation for future research and deployment of educational chatbots, particularly in low-resource languages and policy-rich domains.

II. Related Work

A. Retrieval-Augmented Generation (RAG)

Recent advancements in natural language processing have led to the emergence of RAG frameworks, which integrate information retrieval mechanisms with language generation models to enhance the factual consistency, relevance, and reliability of generated outputs. The RAG model introduced by Lewis et al. [8] demonstrated that combining dense passage retrieval techniques such as Dense Passage Retrieval (DPR) [6], with generative sequence-to-sequence architectures substantially improves performance over traditional retrieval-only or generation-only approaches. By grounding the generative process in dynamically retrieved external evidence, RAG achieved notable success on open-domain question answering benchmarks such as Natural Questions [7]. Further extensions, including REALM [3] and Fusion-in-Decoder [4], have explored enhanced integration of retrieved contexts and improved joint optimization of retrieval and generation objectives. Nevertheless, the application of RAG-based systems in low-resource languages, including Vietnamese, remains limited due to the scarcity of large-scale pretrained retrievers and domain-specific annotated corpora. Although multilingual adaptations [11] and cross-lingual retrieval techniques [15] have been proposed, their effectiveness in specialized domains, such as higher education admissions, remains underexplored, underscoring the need for tailored solutions that address both linguistic and domain-specific challenges.

B. Chunking and Long-Document Processing

Effectively processing long documents remains a central challenge in natural language understanding, particularly for retrieval-augmented and generative systems constrained by fixed input length limitations. Transformer-based architectures,

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such as BERT [2] and RoBERTa [9], are inherently restricted to processing relatively short sequences, typically up to 512 tokens, which necessitates the segmentation of longer documents. A common solution involves dividing documents into smaller, fixed-size chunks; however, naive segmentation often disrupts semantic continuity, fragmenting coherent discourse and thereby impairing retrieval precision and response generation. To mitigate these limitations, several strategies have been explored, including sliding window techniques [10] and context-aware segmentation methods that aim to preserve topical and structural coherence. Maintaining semantic integrity during chunking is particularly critical in retrieval-augmented frameworks, where the relevance of retrieved passages directly impacts the quality and factual grounding of generated outputs. Although architectures such as Longformer [1] have extended the modeling capacity for long sequences through sparse attention mechanisms, their computational demands may render them impractical for domain-specific applications operating under resource constraints. In specialized contexts such as educational admissions, where documents often exhibit complex hierarchical organization, the design of adaptive, context-preserving chunking strategies becomes essential to ensure that each segment remains meaningful and maximally informative for downstream retrieval and reasoning tasks .

III. Methodology

Our system architecture is designed to robustly support university admissions question answering through a modular, multi-stage pipeline. As illustrated in Figure 1, the methodology comprises two principal components: (1) *Contextual Chunking*, which enhances document segmentation by preserving semantic coherence across lengthy, hierarchical texts; and (2) an *Adaptive Query Flow*, a four-stage process encompassing query understanding, historical and contextual expansion, document retrieval, and answer generation. Together, these components enable accurate, contextually grounded, and pedagogically effective responses tailored to the complexities of admissions inquiries.

A. Contextual Chunking

In RAG systems, the accuracy of the retrieval stage is critical to the generation of high-quality responses. This challenge becomes particularly pronounced in domains such as higher education admissions, where documents – including handbooks, policy guidelines, and program catalogs – are lengthy, hierarchically structured, and highly context-dependent. Traditional chunking techniques, which partition documents into fixed-size segments, often disrupt semantic continuity, fragmenting coherent information and thereby impairing retrieval performance.

To address these limitations, we propose Contextual Chunking, a method that dynamically segments documents along semantic and structural boundaries. Building upon conventional segmentation strategies, our approach enriches each chunk with supplementary contextual information derived from the broader document. Specifically, each localized segment is augmented with a concise, document-level summary that encapsulates the overarching theme or intent of the source material. This additional layer of context anchors each chunk within its original semantic framework, thereby mitigating risks of interpretive drift during retrieval. An illustration of the Contextual Chunking process is provided in Figure 2, where document segments (i.e., original chunks) are enriched with semantic summaries (i.e., contextualized chunks) and structural links to preserve coherence for retrieval tasks.

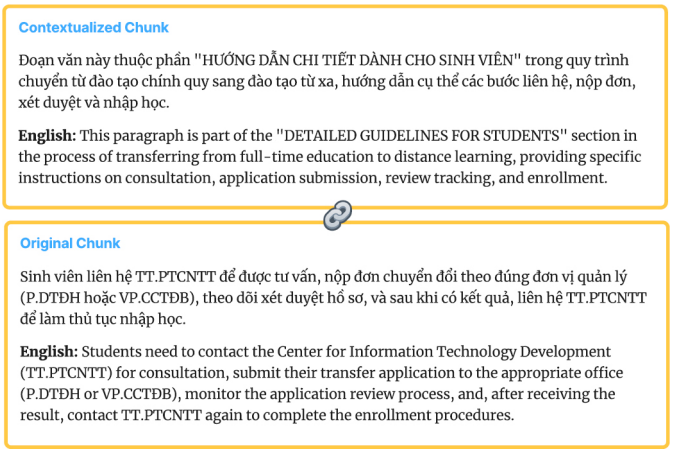

Figure 2. Illustration of the Contextual Chunking process, where document segments are enriched with semantic summaries and structural links to preserve coherence for retrieval.

B. Adaptive Query Flow (AQF)

To address the complex and policy-intensive nature of university admissions inquiries, we propose a modular, four-stage pipeline architecture driven by LLMs. Each stage in this architecture is designed to perform a specialized function, enabling the system to deliver contextually appropriate, accurate, and user-centered responses across diverse admission-related scenarios. This modular design ensures that the system can manage nuanced queries with high precision while maintaining transparency, factual reliability, and scalability. By structuring the process into four interdependent stages, the system effectively addresses challenges in multi-turn dialogue, temporal document alignment, and semantic ambiguity.

1) Domain-Constrained Query Classification: The initial stage of pipeline is dedicated to query classification, serving as a critical mechanism to ensure that user inquiries fall within the domain of university admissions. This process is fundamental given the sensitivity and policy-governed nature of admissions-related information, where maintaining factual accuracy and contextual appropriateness is imperative. Leveraging a LLM, the system conducts a semantic analysis of each incoming query, assessing whether it aligns with predefined admissions categories such as admission quotas, cut-off scores, academic programs, tuition fees, scholarships, admission methods, major offerings, English language requirements, and curriculum structures. Queries that do not correspond to these categories are systematically filtered out,

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and users are informed that their request falls outside the system's operational domain.

2) Dialogue-Aware Contextual Enrichment: The second stage of the pipeline, addresses a fundamental challenge in multi-turn interactions: user queries often rely on implicit assumptions, omit previously stated information, or contain domain-specific abbreviations and shorthand. These linguistic shortcuts introduce ambiguity that can hinder accurate document retrieval. To mitigate this, a dedicated LLM module analyzes the dialogue history to recover semantically relevant context from earlier exchanges and incorporates it into the current query. Furthermore, abbreviations are systematically expanded to match the formal terminology used in structured databases (e.g., converting "UIT" to "University of Information Technology"). This contextual enrichment process significantly enhances query clarity and semantic alignment, thereby improving retrieval precision, reducing misunderstanding, and enabling coherent, human-like interactions across conversational turns.

3) Temporal-Aware Hybrid Retrieval: The document retrieval stage of our system is critical to ensuring the accuracy and relevance of responses in the high-stakes domain of university admissions, where information such as admission quotas, program offerings, tuition structures, and evaluation criteria varies significantly by academic year. To address this temporal sensitivity, we propose a year-aware retrieval mechanism in which each annual dataset is independently indexed and paired with a dedicated retrieval tool. An LLM detects temporal cues in user queries (e.g., references to "2023 admission scores") and dynamically routes retrieval requests to the appropriate year-specific index. This temporal grounding is essential; retrieving documents from an incorrect year could lead to misleading or outdated advice, undermining user trust.

To enhance retrieval fidelity in this stage, we implement a hybrid search strategy that integrates semantic vector search with BM25 keyword-based retrieval, thereby capitalizing on the complementary strengths of each method: while semantic search facilitates the capture of nuanced contextual meanings, BM25 provides robust lexical matching – especially effective for handling domain-specific terminology and named entities prevalent in admissions-related queries. Retrieved documents are reranked using a weighted combination of semantic and lexical scores, improving accuracy and robustness to query phrasing. This ensures contextually relevant and factually accurate responses, critical for high-stakes admissions guidance.

4) Contextually Grounded Response Generation: The final stage of the system, contextually grounded response generation, is critical for ensuring that user-facing outputs are not only informative but also factually reliable – a necessity in high-stakes domains such as university admissions, where precision, transparency, and user trust are paramount. To mitigate the risk of hallucination – particularly in cases where the retrieved context does not explicitly address the query topic – the system enforces a strict constraint whereby the LLM responsible for response generation synthesizes answers exclusively from the retrieved documents. By doing so, it avoids introducing unsupported or fabricated information, thereby ensuring that all outputs remain verifiable, contextually grounded, and aligned with authoritative sources. Moreover, the model is tasked with summarizing and structuring content in a manner that is coherent, pedagogically appropriate, and accessible to a diverse user base.

IV. Vietnamese Admissions QA Dataset (VAQAD)

A. Motivation and Overview

Despite the critical importance of accurate information access in university admissions, no publicly available QA datasets currently exist for the Vietnamese higher education domain. This absence significantly constrains the development, evaluation, and benchmarking of intelligent dialogue systems tailored to this policy-rich and high-stakes context. To bridge this gap, we introduce a specialized, QA dataset designed to facilitate robust system evaluation across dimensions of retrieval effectiveness, contextual reasoning, and answer generation accuracy.

Beyond serving as an evaluation benchmark, the dataset aims to advance research in RAG, domain-specific information retrieval, and multimodal educational AI applications. Its comprehensive design enables fair, transparent, and reproducible comparisons across diverse system architectures, thereby fostering systematic progress in the development of intelligent admissions-support technologies.

B. Dataset Construction

1) Data Sources: To ensure reliability and alignment with current educational policies, the dataset was curated from institutionally validated sources, including official admissions portals, academic brochures, regulatory handbooks, and verified infographics. It incorporates diverse data formats (PDF, DOCX, JPG, PNG) to reflect the heterogeneity of real-world admissions content.

2) Extraction and Annotation: Authentic question–answer (QA) pairs were extracted from inquiries frequently posed by prospective students and matched with authoritative responses issued by the University of Information Technology (UIT) and national regulatory bodies. To preserve semantic fidelity and factual correctness, manual annotation processes were employed throughout data extraction and curation.

3) Preprocessing: Given the heterogeneous and unstructured nature of the collected materials, extensive preprocessing efforts were necessary. Textual normalization, transformation of tabular data into coherent textual form, and semantic enrichment of fragmented content were conducted to ensure consistency.

To facilitate rigorous research and fair system evaluation, we provide both raw and processed versions of the source documents. The raw documents retain their original multimodal and structural characteristics, enabling flexible experimentation with custom preprocessing or retrieval methods. In contrast, the processed documents offer a standardized, high-quality reference for evaluating retrieval and generation tasks. This dual provision is essential for enabling reproducible research,

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comparative analysis, and the development of robust, policy-aligned dialogue systems in the educational domain.

C. Dataset Composition and Statistics

Dataset consists of 54 raw documents comprising 871 pages collected from official university websites. These documents parsed from multiple formats, reflecting the diverse nature of real-world admissions information. From this corpus, we curated a total of 349 QA pairs through manual validation. Each QA pair reflects common user inquiries, derived from authoritative sources and validated against official documents to ensure thematic relevance and factual accuracy.

To support comprehensive evaluation and encourage the development of systems capable of both retrieval and reasoning, each question is annotated with a difficulty label – either easy or hard. Easy questions are those that can be directly answered via factual lookup from a single document, while hard questions require contextual reasoning, synthesis of information across multiple segments, or inference from indirectly related content. Table I summarizes key statistics.

Table I
OVERVIEW OF DATASET STATISTICS

Statistic	Quantity
Admissions and Enrollment	224
Academic Programs and Majors	60
Tuition Fees and Scholarships	25
University Overview and Student Life	40
Total number of questions	349
Hard questions	219 (62.75%)
Easy questions	130 (37.25%)
Average question length	29 tokens
Total number of question tokens	10,234
Average answer length	435 tokens
Total number of answer tokens	151,867

The dataset's balanced composition ensures that evaluation covers a wide range of user intents and query complexities, from direct factual lookups to multi-step reasoning scenarios, thereby providing a robust foundation for advancing context-aware question answering in the domain of higher education admissions.

V. EXPERIMENTS

A. Evaluation Metrics

To ensure a comprehensive and rigorous evaluation of retrieval quality in the absence of ground-truth document chunks, we adopt a multi-faceted metric framework encompassing both traditional and context-aware measures with arguments: input, actual output, expected output and retrieval output. Specifically, we utilize five metrics:

Mean Reciprocal Rank (MRR) reflects how early the first relevant document appears, highlighting ranking quality.

Recall@K measures the fraction of relevant items found among the top-K results, reflecting retrieval coverage.

Contextual Precision (ConPre)[1] evaluates how effectively the system prioritizes content identified by an LLM-as-a-judge as relevant. It reflects ranking quality by rewarding early placement of useful content.

Contextual Recall (ConRec)[2] measures how well the retrieved context supports the reference answer, based on overlap with key answer components.

Contextual Relevancy (ConRel)[3] quantifies the overall pertinence of retrieved content to the query, regardless of the reference answer. It reflects general contextual alignment as assessed by an LLM-as-a-judge. A fixed threshold $\tau = 0.7$ is applied across all contextual metrics.

B. Contextual Chunking

To evaluate the impact of contextual chunking on system performance, we conducted a comparative analysis using five evaluation metrics, as shown in Table II. Contextual chunking enhances retrieval performance across multiple dimensions, with notable improvements in Mean Reciprocal Rank (from 70.3% to 78.2%), Recall5 (from 81.2% to 88.9%), and Contextual Relevancy (from 29.51% to 34.96%) compared to the non-contextual setting. These gains reflect improved ranking quality and stronger alignment between retrieved content and user intent, particularly in scenarios where semantic continuity is essential. Although Contextual Precision and Contextual Recall show marginal declines (approximately 1% and 0.5% respectively), this is likely due to the richer semantic scope of contextual chunks, which, while more informative, may introduce peripheral details not strictly necessary for answering the query. Nonetheless, the overall improvements affirm the effectiveness of context-aware chunking in enhancing retrieval accuracy and robustness, especially for complex and policy-heavy university admissions queries.

C. Adaptive Query Flow

To assess the overall effectiveness of our system, we conducted a series of experiments utilizing a multi-LLM architecture under two distinct chunking strategies: Non-Contextual and the proposed Contextual Chunking. The evaluation focused on end-to-end system accuracy, with results summarized in Table III to demonstrate the performance impact of each approach.

The results demonstrate that contextual chunking significantly boosts overall system accuracy, improving from 74.21% to 87.1%, with a notable increase in correct responses to hard questions from 163 to 190. This improvement highlights the method's effectiveness in preserving semantic structure, which enhances the retrieval of relevant context for complex, multi-faceted queries. Furthermore, the observed performance gains validate the design of the four-stage LLM architecture, where each specialized module, responsible for query understanding, contextual enrichment, document retrieval, and response generation, works in concert to deliver accurate

[1] https://deepeval.com/docs/metrics-contextual-precision
[2] https://deepeval.com/docs/metrics-contextual-recall
[3] https://deepeval.com/docs/metrics-contextual-relevancy

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Table II
RETRIEVAL PERFORMANCE COMPARISON BETWEEN DIFFERENT CHUNKING STRATEGIES.

Setting	MRR (%)	Recall@5 (%)	ConPre (%)	ConRec (%)	ConRel (%)
Non-Contextual Chunking	70.3	81.2	**65.9**	**71.06**	29.51
Contextual Chunking	**78.2**	**88.9**	64.76	70.49	**34.96**

Table III
EVALUATION RESULTS ON THE PROPOSED DATASET.

Setting	Multi-LLM + Non-Contextual Chunking	Multi-LLM + Contextual Chunking
Passed Easy question	96	114
Passed Hard question	163	190
Correctness	259	**304**
Accuracy	74.21	**87.1**

and contextually grounded answers, especially in high-stakes domains like university admissions.

VI. CONCLUSION

In this work, we present a framework for advancing university admissions question answering in Vietnamese through three core contributions: a domain-specific dataset curated from validated institutional sources; a contextual chunking framework that preserves semantic coherence in long and heterogeneous documents; and a modular multi-LLM-based architecture that improves scalability, transparency, and task specialization. Collectively, these contributions address key limitations of traditional rule-based and retrieval-based systems by enhancing retrieval accuracy, contextual understanding, and system modularity. This work establishes a foundation for future research and the practical deployment of intelligent educational chatbots in low-resource, policy-rich environments, contributing to the objective of improving accessibility and transparency in higher education admissions. Despite strong performance, the system faces deployment challenges like multi-stage latency and yearly index maintenance, requiring further work for scalable QA applications.

ACKNOWLEDGMENT

This research is funded by University of Information Technology-Vietnam National University HoChiMinh City under grant number D1-2025-23

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